

ELECTRICAL INSTRUMENTATION: PHASE 3

Advanced Piezoelectric Transducer Systems

Grassroots Derivations, Charge Amplification, and Accelerometer Dynamics

Semester Preparation Masterclass

Contents

1	Grassroots Piezoelectric Physics	2
1.1	The 'd' and 'g' Sensitivity Constants	2
2	Equivalent Circuit & The Dynamic Limitation	4
2.1	The Parallel Electro-Mechanical Topology	4
2.2	Derivation of the High-Pass Transfer Function	5
2.3	Why Steady State Gain is Mathematically Zero	5
3	The Charge Amplifier Imperative	7
3.1	The Cable Capacitance Catastrophe	7
3.2	The Virtual Ground Solution	7
3.3	The Mandatory Feedback Resistor (R_f)	8
4	The Piezoelectric Accelerometer	9
4.1	Physical Construction and Concept	9
4.2	Block-by-Block Full Transfer Function Derivation	10
4.3	Engineering Bandwidth Justification	11

Grassroots Piezoelectric Physics

Piezoelectric transducers are active sensors that generate an electrical charge when subjected to mechanical stress. They operate on the *direct piezoelectric effect*, where physical deformation of the crystalline lattice disrupts the centers of positive and negative charge, polarizing the material.

The 'd' and 'g' Sensitivity Constants

To engineer a piezoelectric system, one must command the two fundamental sensitivity constants that bridge the mechanical and electrical domains.

1. The Charge Sensitivity Constant (d): This constant dictates the fundamental charge generation capability of the material. It is defined as the total electrical charge (q) generated strictly per unit of applied mechanical force (F).

$$d = \frac{q}{F} \quad [\text{Coulombs} / \text{Newton (C/N)}] \quad (1)$$

2. The Voltage Sensitivity Constant (g): This constant defines the resulting internal electric field intensity (E) generated strictly per unit of applied mechanical stress (P).

$$g = \frac{E}{P} \quad [\text{Volt-meters} / \text{Newton (V}\cdot\text{m/N)}] \quad (2)$$

Mathematical Proof of the $g - d$ Relationship

Let the piezoelectric crystal have a total thickness t and a cross-sectional surface area A . The mechanical stress P is force per unit area: $P = \frac{F}{A}$. The electric field intensity E across parallel surfaces is the generated voltage V divided by the thickness t : $E = \frac{V}{t}$.

Substituting these definitions into the g constant equation:

$$g = \frac{\left(\frac{V}{t}\right)}{\left(\frac{F}{A}\right)} = \frac{V \cdot A}{F \cdot t} \quad (3)$$

The physical geometry of the crystal inherently forms a parallel-plate capacitor. The fundamental capacitance is $C = \frac{\epsilon A}{t}$, where ϵ is the absolute permittivity of the crystal

material. From fundamental electrostatic charge laws, $V = \frac{q}{C}$.

Substituting this voltage definition into our equation:

$$g = \frac{\left(\frac{q}{C}\right) \cdot A}{F \cdot t} = \frac{q \cdot A}{C \cdot F \cdot t} \quad (4)$$

Now, substitute the geometric capacitance formula $C = \frac{\epsilon A}{t}$ directly into the denominator:

$$g = \frac{q \cdot A}{\left(\frac{\epsilon A}{t}\right) \cdot F \cdot t} \quad (5)$$

The geometric terms A and t mathematically cancel out entirely, leaving:

$$g = \frac{q}{\epsilon \cdot F} \quad (6)$$

Since we previously defined the charge sensitivity $d = \frac{q}{F}$, we substitute d to arrive at the absolute final relation:

$$g = \frac{d}{\epsilon} \implies d = \epsilon \cdot g \quad (7)$$

This derivation rigorously proves that the relationship between charge sensitivity and voltage sensitivity is dictated exclusively by the **dielectric permittivity** (ϵ) of the material, not the magnetic permeability.

Equivalent Circuit & The Dynamic Limitation

A piezoelectric crystal cannot be analyzed in a vacuum. It must be connected to an electronic amplifier via a cable. This creates a complex parallel electrical topology.

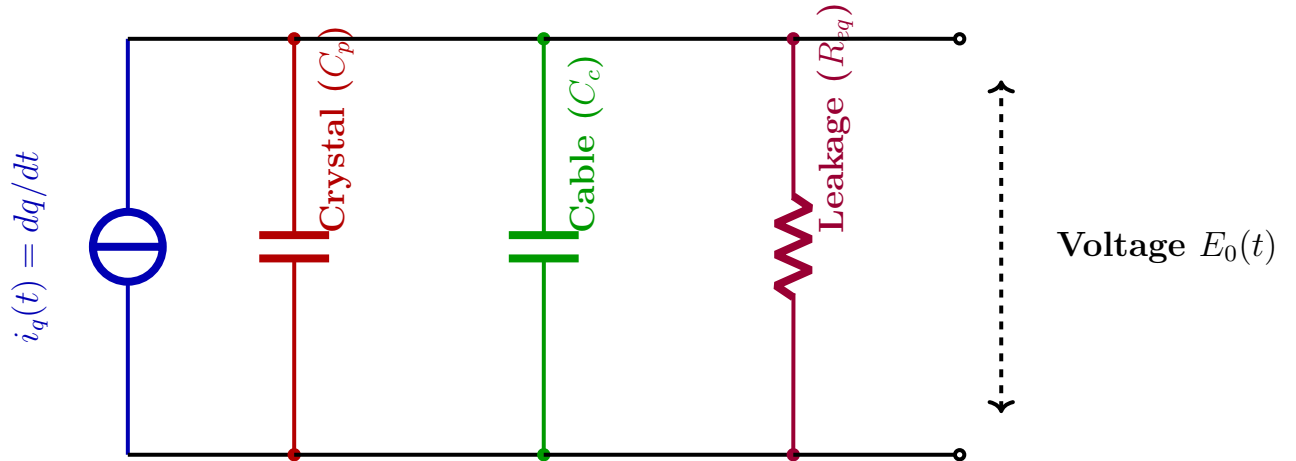
The Parallel Electro-Mechanical Topology

By utilizing Norton's equivalent circuit theorem, the piezoelectric crystal is modeled purely as an ideal dynamic current source:

$$i_q(t) = \frac{dq(t)}{dt} \quad (8)$$

This current source is placed in parallel with the crystal's intrinsic internal capacitance C_p and its internal leakage resistance R_p .

When interfaced with a voltage amplifier (presenting input resistance R_a and input capacitance C_a) via a coaxial cable (introducing shunt capacitance C_c), the entire system collapses into a massive parallel network.



The total equivalent parallel capacitance (C_{eq}) is the sum of all capacitive elements:

$$C_{eq} = C_p + C_c + C_a \quad (9)$$

The total equivalent parallel resistance (R_{eq}) governs the discharge rate:

$$R_{eq} = \frac{R_p \cdot R_a}{R_p + R_a} \quad (10)$$

The critical electrical time constant (τ) is the product:

$$\tau = R_{eq} \cdot C_{eq} \quad (11)$$

Derivation of the High-Pass Transfer Function

Using Laplace transforms, the total equivalent impedance of the parallel network is:

$$Z(s) = \frac{R_{eq} \cdot \left(\frac{1}{sC_{eq}}\right)}{R_{eq} + \left(\frac{1}{sC_{eq}}\right)} = \frac{R_{eq}}{1 + sR_{eq}C_{eq}} \quad (12)$$

The output voltage $E_0(s)$ developed across this impedance is the product of the dynamic current $I_q(s)$ and the impedance $Z(s)$. Since $i_q = \frac{dq}{dt}$, then in the s-domain, $I_q(s) = sQ(s)$.

$$E_0(s) = I_q(s) \cdot Z(s) = sQ(s) \left[\frac{R_{eq}}{1 + sR_{eq}C_{eq}} \right] \quad (13)$$

Substitute the system time constant $\tau = R_{eq}C_{eq}$. We can rewrite R_{eq} as $R_{eq} = \frac{\tau}{C_{eq}}$:

$$E_0(s) = Q(s) \left[\frac{s \left(\frac{\tau}{C_{eq}}\right)}{1 + s\tau} \right] = \frac{Q(s)}{C_{eq}} \left[\frac{s\tau}{1 + s\tau} \right] \quad (14)$$

If the system operates as a mechanical displacement sensor, the charge is proportional to the input displacement: $Q(s) = K_q X_i(s)$. The final transfer function becomes:

$$\frac{E_0(s)}{X_i(s)} = \left(\frac{K_q}{C_p + C_c + C_a} \right) \left[\frac{s\tau}{1 + s\tau} \right] \quad (15)$$

Why Steady State Gain is Mathematically Zero

A common trap in BJB Sir's exams is asking if a piezoelectric system possesses steady-state gain or can measure static force.

A "steady-state" or static input implies that the frequency of the mechanical input is

exactly zero ($\omega = 0$, or $s = 0$). Evaluating the magnitude of our derived high-pass transfer function at steady-state ($\omega = 0$):

$$|G(j0)| = \left(\frac{K_q}{C_{eq}} \right) \frac{0 \cdot \tau}{\sqrt{1 + 0}} = 0 \text{ Volts} \quad (16)$$

Physical Reality: When a static, continuous force is applied, a charge packet $+q$ is generated instantly. However, because R_{eq} is finite, it provides a closed electrical path for the charge to physically bleed away to ground. The output voltage decays exponentially to zero ($E_0 \propto e^{-t/\tau}$) despite the heavy physical load remaining clamped on the crystal. Therefore, piezoelectric systems are strictly dynamic-only instruments and **can never possess a steady-state gain**.

The Charge Amplifier Imperative

The Cable Capacitance Catastrophe

Examine the constant multiplier in our derived transfer function: $\left(\frac{K_q}{C_p + C_c + C_a}\right)$. The cable capacitance (C_c) sits aggressively in the denominator. Coaxial cables are highly unstable. If the cable vibrates, bends, or is subjected to thermal gradients, its physical internal geometry shifts, causing C_c to violently fluctuate. Because C_c fluctuates, the denominator fluctuates, causing the measured voltage to spike wildly. This generates massive triboelectric noise voltages that completely swamp and destroy the actual transducer signal. Voltage amplifiers are practically useless for precision piezoelectric measurements.

The Virtual Ground Solution

To resolve this absolute failure, engineers replace the standard voltage amplifier with a **Charge Amplifier**. A Charge Amplifier utilizes an Operational Amplifier (Op-Amp) with a highly stable, precision feedback capacitor (C_f) placed directly across it.

Due to the Op-Amp's near-infinite open-loop gain, the inverting input terminal is forcefully clamped to exactly 0 Volts (the **Virtual Ground** principle). **If the input node is anchored at 0V, absolutely no voltage can ever develop across the unstable cable capacitance C_c .** From electrostatics ($Q = C \times V$), if the voltage across the cable is zero, the cable stores absolutely zero stray charge ($Q = C_c \times 0 = 0$).

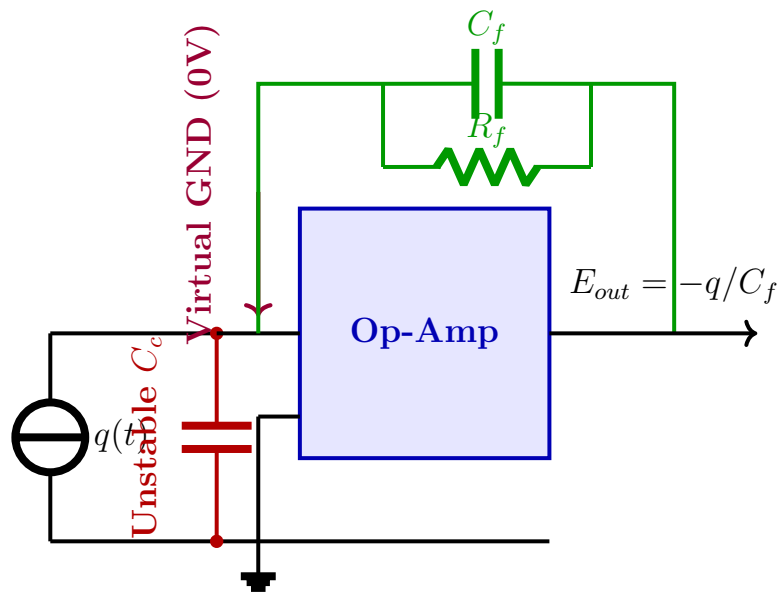
Therefore, the entire pristine charge packet q generated by the crystal is forcefully swept entirely into the highly stable internal feedback capacitor C_f . The output voltage of the Op-Amp becomes strictly dependent on the feedback loop:

$$E_0 = -\frac{q}{C_f} \tag{17}$$

By utilizing the Virtual Ground, the system achieves total mathematical immunity to coaxial cable length and bending noise.

The Mandatory Feedback Resistor (R_f)

A theoretical charge amplifier uses only a capacitor (C_f). However, a practical charge amplifier **must** include a massive resistor ($R_f \approx 10^{11}\Omega$) in parallel with C_f . Real Op-Amps possess microscopic input bias currents. If only a capacitor is present, these bias currents will continuously pump charge into C_f , ramping the output voltage until it slams violently into the power supply rails (saturation), paralyzing the amplifier. The parallel resistor R_f provides a mandatory DC leakage path, bleeding off the bias currents and maintaining the Op-Amp stably at 0V DC.

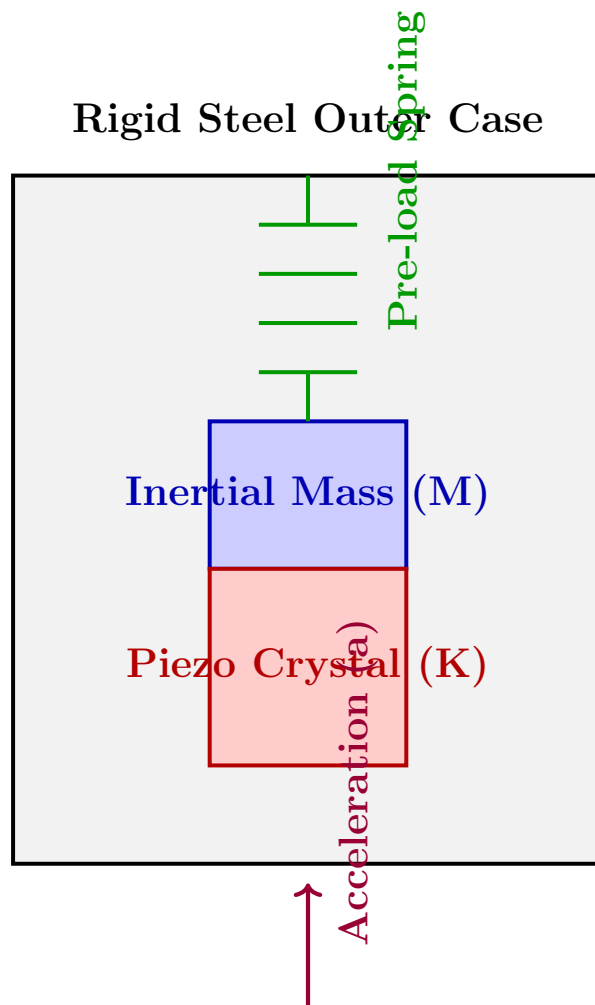


The Piezoelectric Accelerometer

Physical Construction and Concept

A piezoelectric accelerometer does not measure acceleration by sensing space and time. It employs Newton's Second Law of Motion ($F = m \cdot a$) in a rigid mechanical housing. A known, calibrated internal seismic mass (M) rests directly atop a stiff piezoelectric crystal block. The entire assembly is mechanically pre-loaded under heavy compression by a stiff spring. This ensures the mass never physically separates from the crystal during violent upward decelerations, as piezoelectric ceramics fracture easily under tension.

When the rigid outer housing experiences an external vibration or acceleration ($a = \ddot{x}_i$), inertia dictates that the internal mass resists the change in velocity. This exerts a dynamic compressive force $F_{dynamic} = M \cdot \ddot{x}_i$ down onto the crystal structure.



Block-by-Block Full Transfer Function Derivation

A piezoelectric accelerometer operates as a coupled electro-mechanical system. It requires the sequential multiplication of two distinct physical transfer functions: the Mechanical mass-spring system, and the Electrical charge amplifier.

Part 1: The Mechanical Subsystem (Seismic Mass) Let the external vibration input displacement be $x_i(t)$. Let the absolute displacement of the mass be $x_m(t)$. The relative displacement (the actual compression of the crystal) is $x_0(t) = x_m(t) - x_i(t)$.

By Newton's Second Law, the sum of forces on the mass is governed by the spring (K) and damper (B) opposing its motion:

$$M\ddot{x}_m = -B\dot{x}_0 - Kx_0 \quad (18)$$

Substitute $x_m = x_0 + x_i$ and its second derivative:

$$M(\ddot{x}_0 + \ddot{x}_i) = -B\dot{x}_0 - Kx_0 \implies M\ddot{x}_0 + B\dot{x}_0 + Kx_0 = -M\ddot{x}_i \quad (19)$$

Take the Laplace transform (assuming zero initial conditions):

$$(Ms^2 + Bs + K)X_0(s) = -Ms^2X_i(s) \quad (20)$$

We define the input variable strictly as acceleration $A(s) = s^2X_i(s)$. Therefore, the mechanical transfer function is:

$$\frac{X_0(s)}{A(s)} = \frac{-M}{Ms^2 + Bs + K} \quad (21)$$

Divide numerator and denominator by the stiffness K , and substitute the natural frequency $\omega_n^2 = K/M$ and damping ratio $2\zeta/\omega_n = B/K$:

$$\frac{X_0(s)}{A(s)} = \frac{-1/\omega_n^2}{(s/\omega_n)^2 + 2\zeta(s/\omega_n) + 1} \quad (22)$$

Part 2: The Electrical Subsystem (Charge Amplifier) The mechanical compression $X_0(s)$ generates an electrical charge proportional to the crystal's sensitivity K_q :

$$Q(s) = K_q \cdot X_0(s) \quad (23)$$

To measure this charge without cable capacitance errors, it is fed into a Charge Amplifier characterized by a feedback capacitor C_f and a mandatory parallel feedback resistor R_f . The transfer function of a standard inverting charge amplifier is a pure high-pass filter defined by time constant $\tau = R_f C_f$:

$$\frac{E_0(s)}{Q(s)} = \frac{-1}{C_f} \left[\frac{s\tau}{1 + s\tau} \right] \quad (24)$$

Part 3: The Unified Transfer Function The total system transfer function is the strict mathematical product of the mechanical and electrical blocks:

$$G_{total}(s) = \frac{E_0(s)}{A(s)} = \left[\frac{X_0(s)}{A(s)} \right] \times \left[\frac{Q(s)}{X_0(s)} \right] \times \left[\frac{E_0(s)}{Q(s)} \right] \quad (25)$$

Substitute the derived expressions. The two negative signs mathematically cancel out:

$$\frac{E_0(s)}{A(s)} = \left[\frac{K_q}{\omega_n^2 \cdot C_f} \right] \left[\frac{s\tau}{1 + s\tau} \right] \left[\frac{1}{(s/\omega_n)^2 + 2\zeta(s/\omega_n) + 1} \right] \quad (26)$$

Engineering Bandwidth Justification

This masterful equation proves the piezoelectric accelerometer acts as a dynamic band-pass filter. A sensor is only useful in its **flat frequency band** where the magnitude $|G(j\omega)|$ is constant.

1. The Lower Limit (Electrical High-Pass): At very low frequencies, the term $\frac{\omega\tau}{\sqrt{1+(\omega\tau)^2}}$ causes severe signal attenuation. For a maximum allowable measurement error of 5%, the magnitude must remain ≥ 0.95 :

$$0.95 = \frac{\omega\tau}{\sqrt{1+(\omega\tau)^2}} \implies (\omega\tau)^2 = 0.9025(1 + (\omega\tau)^2) \quad (27)$$

Solving for $\omega\tau$ yields $\omega\tau \approx 3.04$. Thus, the lower cut-off frequency is bounded at:

$$\omega_{lower} = \frac{3.04}{\tau} \quad (28)$$

2. The Upper Limit (Mechanical Resonance): At high frequencies, the mechanical 2nd-order low-pass term peaks violently near resonance ($\omega = \omega_n$). To maintain a flat response within $\sim 5\%$ error without triggering catastrophic resonant peaking (assuming

nominal damping $\zeta \approx 0.6$), mathematical analysis restricts safe operation to less than 20% of the natural frequency.

$$\omega_{upper} = \frac{\omega_n}{5} \quad (29)$$

Final Bandwidth Proof: The absolute usable frequency bandwidth for linear measurement is rigorously bounded by:

$$\frac{3.04}{\tau} \leq \omega \leq \frac{\omega_n}{5} \quad (30)$$